



Firm-Specific News Sentiment and Short-Horizon Risk-Adjusted Return Predictability: Evidence from Large Language Models

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01

Introduction

Motivation, problem statement, and research questions

Research Context & Motivation

Narrative-Driven Markets

- Financial markets are increasingly shaped by narratives, disclosures, and digital communication.
- Prices reflect heterogeneous expectations ("investor sentiment").
- Modern NLP enables fine-grained sentiment extraction.

The Central Gap

- Most studies use document-level sentiment.
- One article often discusses multiple firms.
- Firm-specific signals become diluted.

Example:

"Netflix shares dip despite Apple's surge in streaming subscribers."

 Document-level sentiment → Neutral

 Target-level sentiment:

- Netflix → Negative
- Apple → Positive

The Thesis Answer

- ✓ Target-level sentiment extraction
- ✓ Benchmarking modern LLMs
- ✓ Economic validation through asset pricing

Problem Statement



LLMs have greatly improved financial sentiment extraction.



Their economic value remains largely unexplored.



It is unclear whether firm-level narrative sentiment improves:

- short-horizon return prediction
- risk-adjusted abnormal returns



This thesis evaluates both predictive and economic value.



Research Questions

RQ1 How accurately do LLMs extract firm-level sentiment compared with lexicon-based and transformer models?

RQ2 Does few-shot prompting improve target-level financial sentiment analysis?

RQ3 Are firm-level sentiment signals associated with short-horizon stock returns?

RQ4 How do textual depth (headlines vs. articles) and predictive models (linear vs. nonlinear) affect forecasting accuracy?

RQ5 Does LLM-derived sentiment provide incremental predictive value and generate risk-adjusted alpha?

Three Primary Contributions

1. Measurement Infrastructure

- Two manually annotated datasets
 - 1476 headlines
 - 1170 full-length articles
- High inter-annotator reliability
 - Krippendorff's $\alpha > 0.81$
- Publicly released* for reproducibility

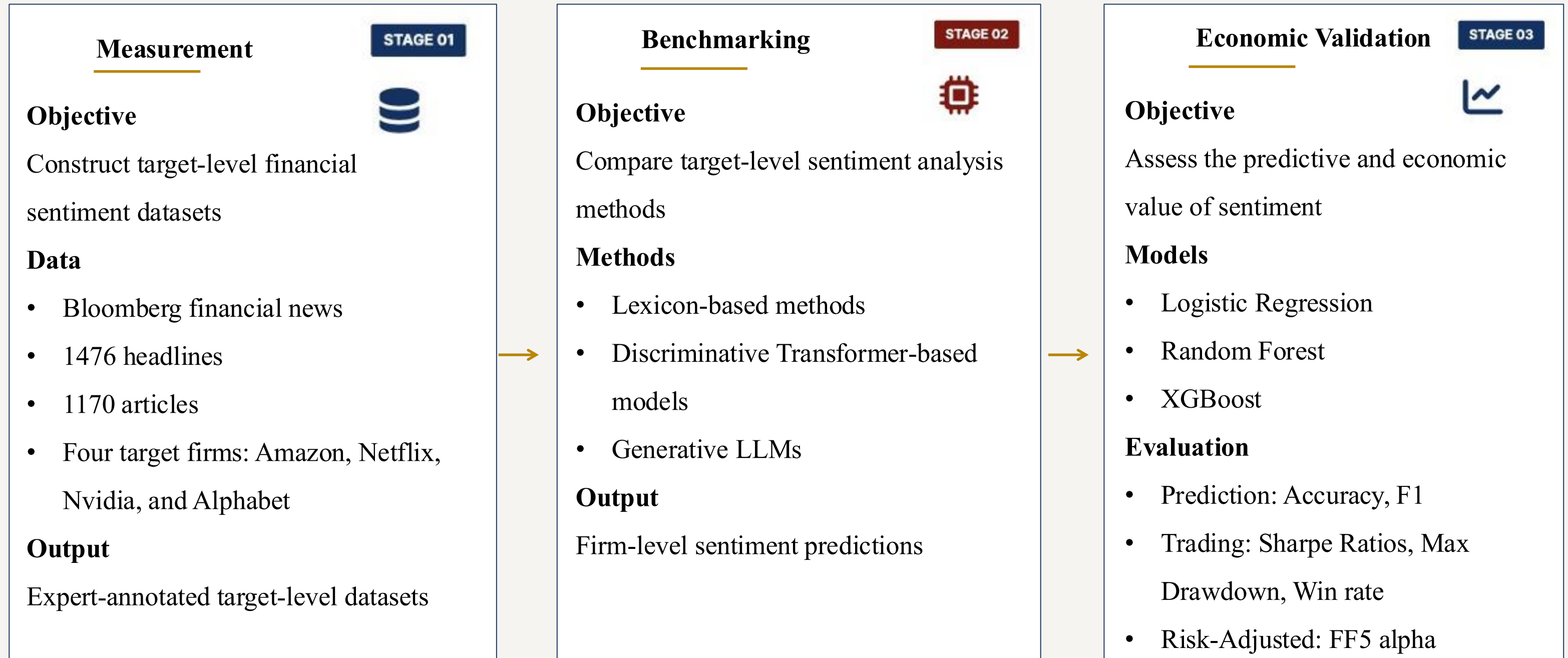
2. Methodological Benchmarking

- Lexicon-based methods
- Discriminative Transformer models
- Generative LLMs

3. Economic Validation

- Short-horizon return prediction
- Linear & nonlinear prediction models
- Fama-French 5-factor alpha

Research Design: Three-Stage Framework



02

Literature Review

Sentiment measurement methods, textual sentiment analysis, granularity issues

Measuring Investor Sentiment: A Taxonomy

Market-Implied

- Trading volume
- IPO volume
- First-day returns

Internet Query

- Google Trends
- FEARS index (Financial and Economic Attitudes Revealed by Search)

Survey-Based

- AAI Sentiment Survey
- Michigan Consumer Sentiment
- Investor Intelligence Survey

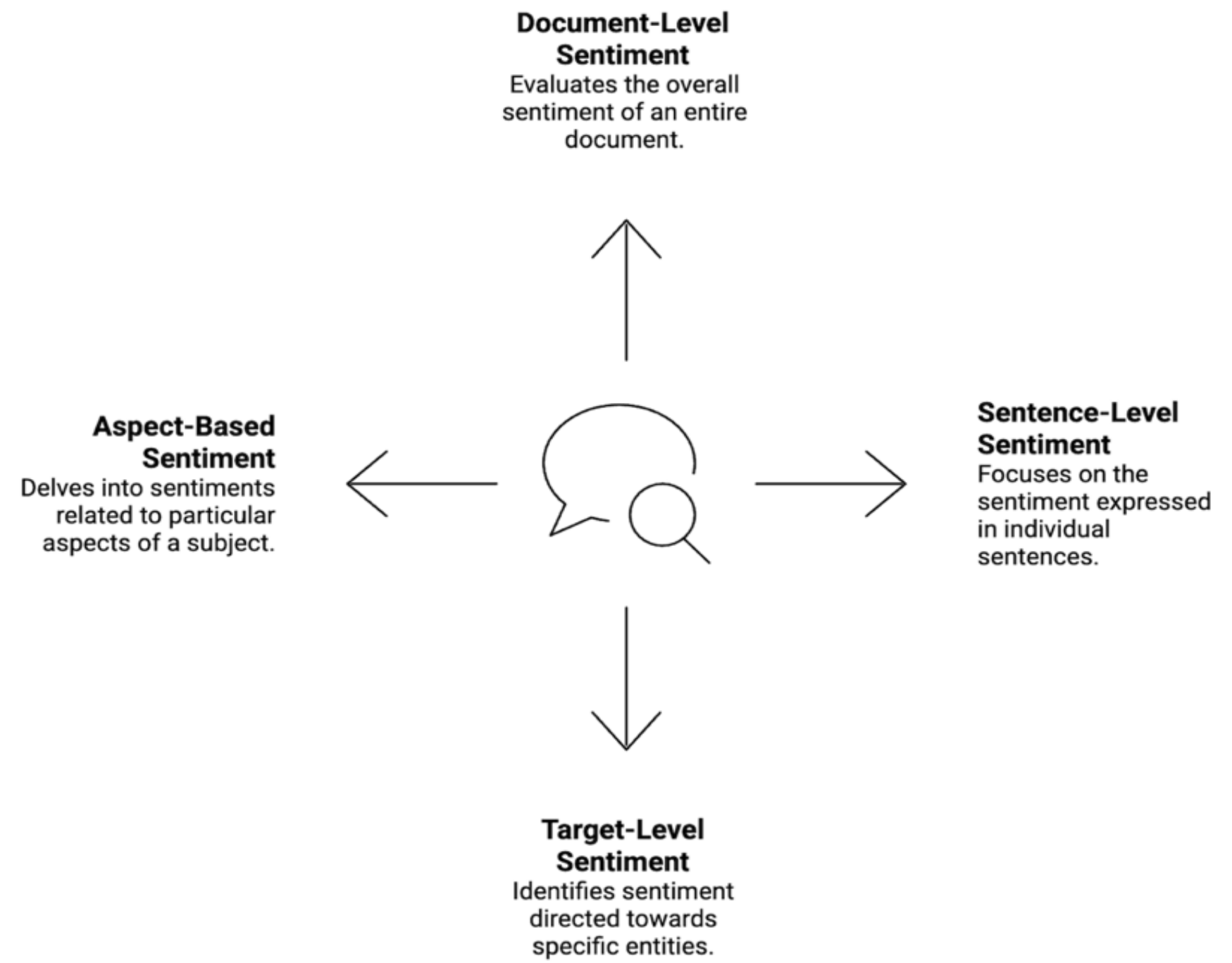
Textual Sentiment (This Thesis)

- News articles
- Social media
- Corporate disclosure
- Earnings calls.

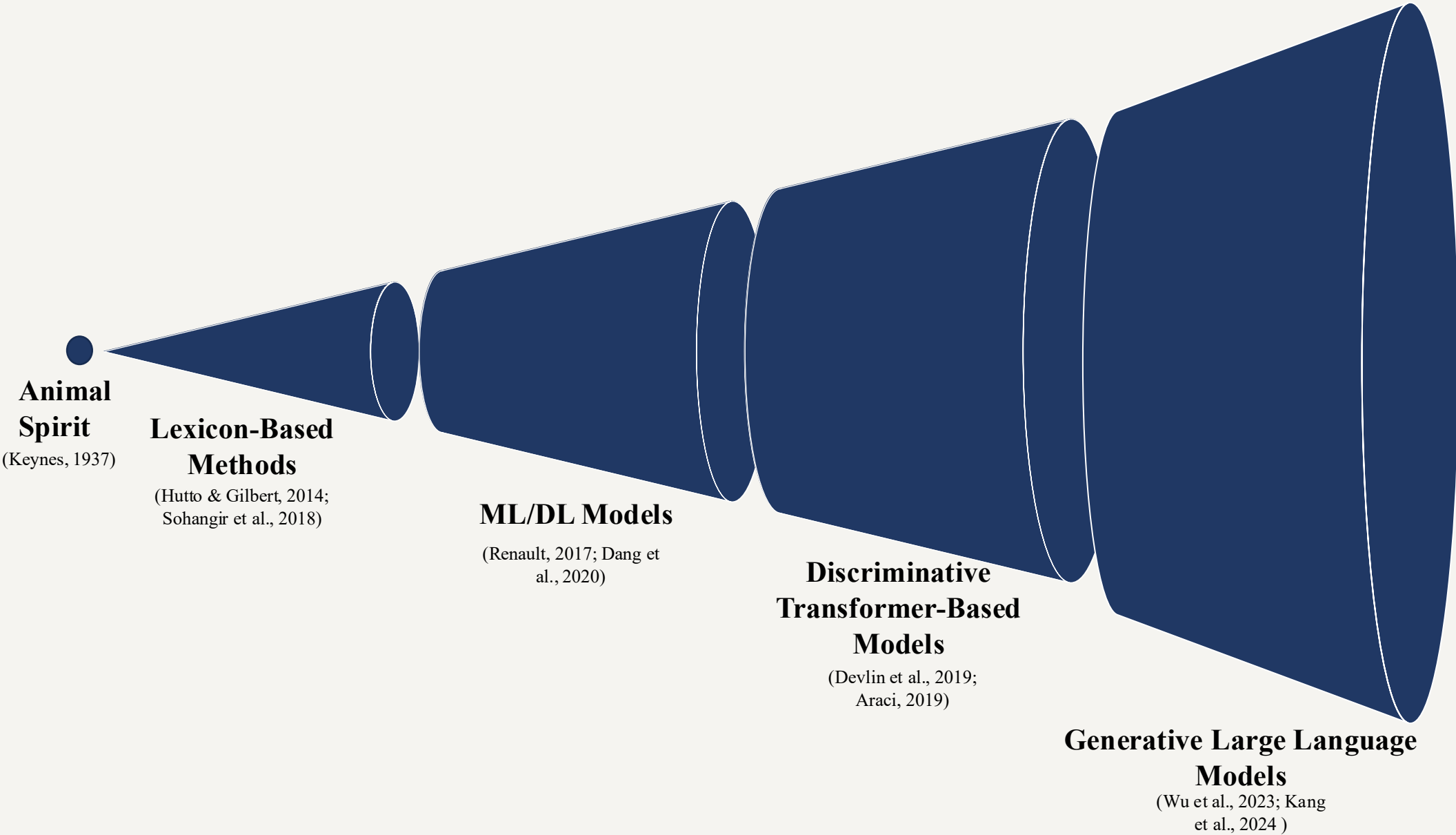
Unconventional

- Weather conditions
- Sports outcomes
- Disasters
- Terrorist attacks

Textual Sentiment Granularity



Evolution of Textual Sentiment Methods



Research Gaps

1. Target-Level Data Scarcity

- Limited expert-annotated firm-level sentiment datasets

2. Methodological Benchmarking

Limited systematic comparison across:

- Lexicon-based methods
- Discriminative Transformer-based models
- Generative LLMs

3. Informational Depth

- Limited evidence on the predictive value of full-length articles relative to headlines.

4. Risk-Adjusted Validation

- Most studies focus on statistical accuracy
- Limited evidence on risk-adjusted performance

03

Target-Level Financial Sentiment Analysis: Headlines

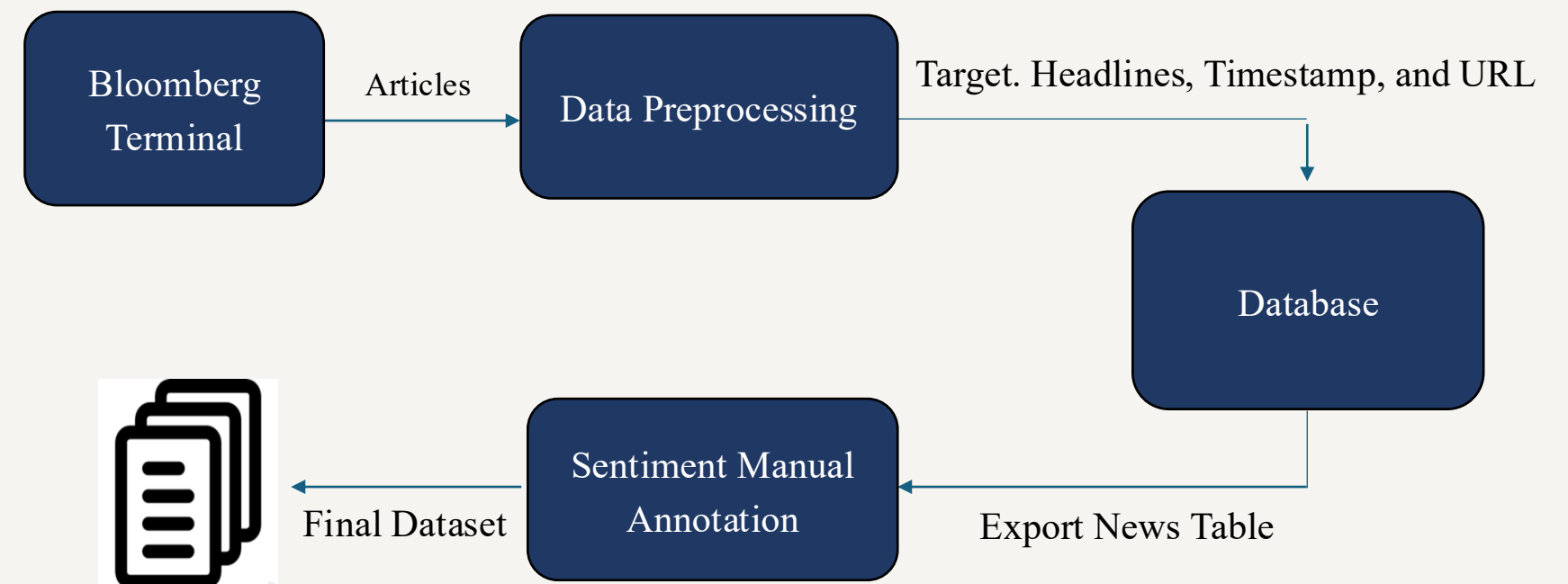
Dataset construction, annotation methodology, and benchmarking results

On Assessing the Performance of LLMs for Target-Level Sentiment Analysis in Financial News Headlines

Published in *Algorithms* (2025)

Headline Dataset Construction

- **Source:** Bloomberg Terminal
- **Headlines:** 1476
- **Period:** Sept 2023 - Jan 2024
- **Firms:** Amazon, Netflix, Nvidia, Alphabet
- **Annotators:** 3 expert annotators
- **Reliability:** Krippendorff's Alpha = 0.82
- **Classes:** Positive (+1), Negative (-1), Neutral (0)
- **Publicly released** for reproducibility



Dataset creation workflow

Dataset Statistics and Annotation Examples

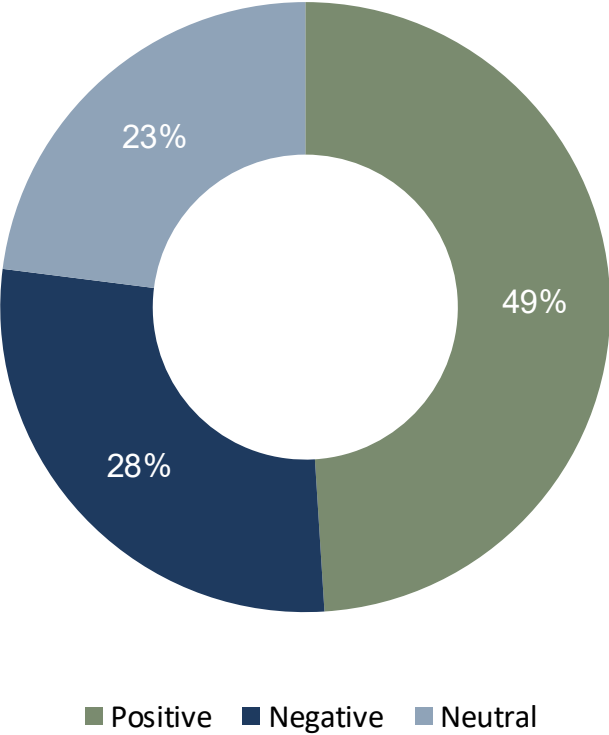
Data Statistics

Target	Headlines	Daily Avg. (SD)
Amazon	573	5.07 (4.41)
Netflix	298	3.77 (5.14)
Nvidia	392	3.70 (4.28)
Alphabet	213	3.04 (3.95)
Total	1476	15.58 (17.78)

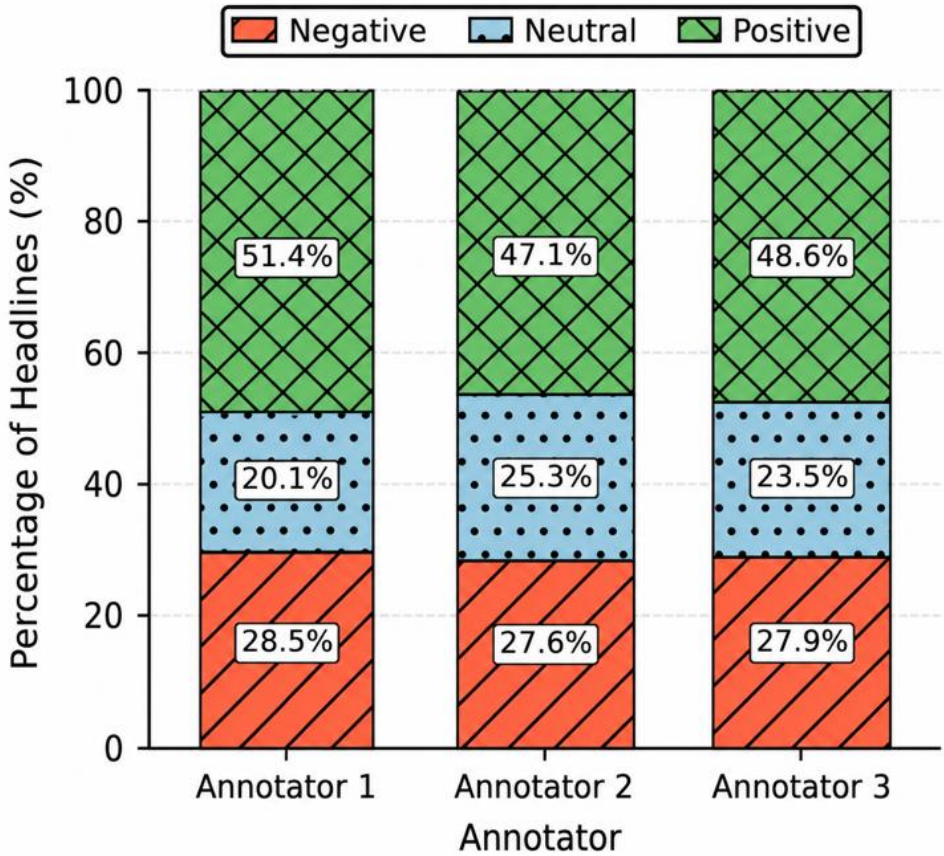
Annotation Examples

Target	Headline	Label
Amazon	Pioneer Disciplined Growth Buys More Nike Class B, Cuts Amazon	-1
Netflix	Netflix CFO Speaks at Bank of America Conference	0
Nvidia	Nvidia to Partner with India's Tata, Reliance for AI	1
Alphabet	Alphabet Hits 52-Week High at \$139.17	1

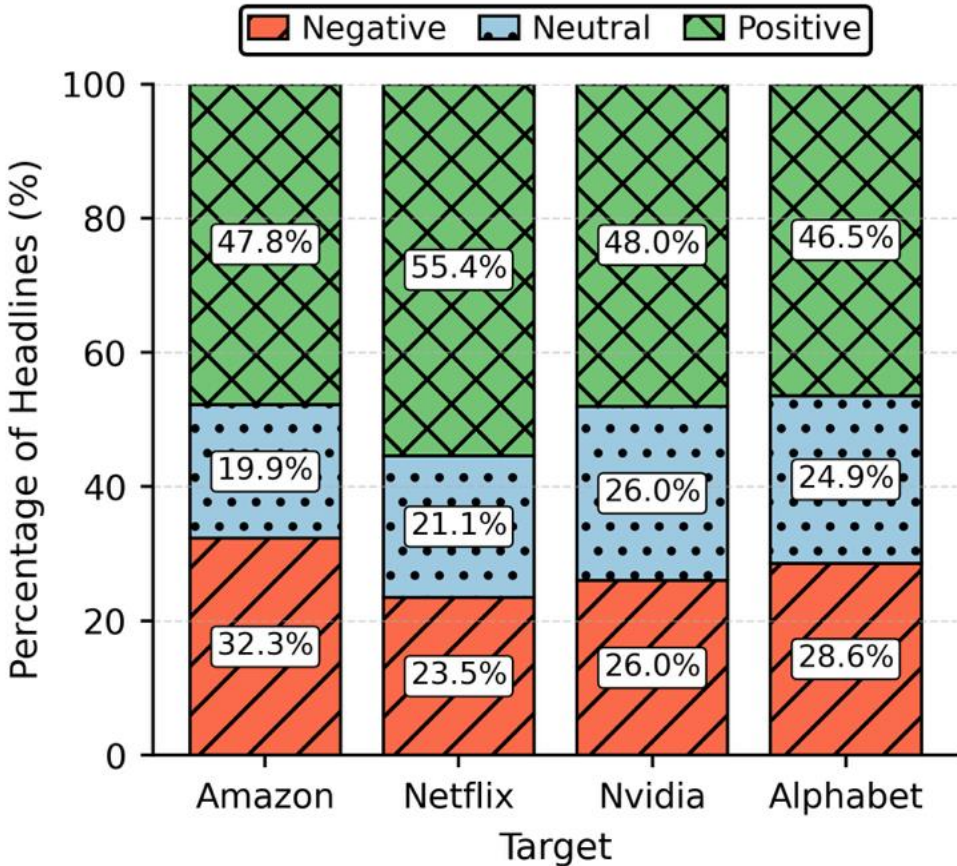
Sentiment Distribution



Sentiment distribution in the overall dataset



Annotator-assigned sentiment distribution



Sentiment distribution across targets

Models Benchmarked

Rule-Based & Discriminative Transformer-Based Models



- **Rule-Based**
 - VADER
- **Discriminative Transformer-Based**
 - DistilFinRoBERTa
 - DeBERTa-v3-base-ABSA

Generative LLMs



- ChatGPT (3.5, 4, and 4o)
- Gemini (1 Pro, 1.5 Flash, 1.5 Pro)



Zero-Shot Prompt

"Evaluate the sentiment conveyed by the headline toward the {target} from an investor's perspective. Classify the sentiment as Positive (1), Negative (-1), or Neutral (0)."

Performance Outcomes

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted Precision	Weighted Recall	Weighted F1
VADER	0.49	0.51	0.47	0.48	0.53	0.49	0.50
DistilFinRoBERTa	0.51	0.62	0.54	0.50	0.68	0.51	0.53
DeBERTa-v3 (pre)	0.52	0.67	0.58	0.54	0.75	0.52	0.53
DeBERTa-v3 (ft)	0.87	0.85	0.84	0.84	0.87	0.87	0.87
ChatGPT-3.5	0.75	0.77	0.75	0.74	0.82	0.75	0.77
ChatGPT-4	0.86	0.84	0.83	0.83	0.86	0.86	0.86
ChatGPT-4o	0.84	0.82	0.83	0.82	0.86	0.84	0.84
Gemini 1 Pro	0.78	0.75	0.73	0.74	0.77	0.78	0.77
Gemini 1.5 Pro	0.83	0.82	0.83	0.81	0.86	0.83	0.84
Gemini 1.5 Flash	0.83	0.80	0.80	0.80	0.83	0.83	0.83
Note: ft = fine-tuned; pre = pre-trained							

Key Findings & Conclusion

- Tradition NLP models struggle with target-level sentiment
- Fine-tuned DeBERTa-v3 achieved the highest performance
- Generative LLMs achieve competitive performance **without any task-specific fine-tuning**

Conclusion: LLMs as Viable Off-the-Shelf Solutions

- Competitive performance without task-specific fine-tuning
- Reduced dependence on resource-intensive manual annotation and model adaptation
- Practical and scalable solution for target-level financial sentiment analysis

04

Target-Level Financial Sentiment Analysis: Articles

Full-length articles, few-shot evaluation, and financial alignment

Preliminary Version

Benchmarking Large Language Models
for Target-Based Sentiment Analysis

Published in *CliC-it 2025*

Extended Version

Benchmarking Large Language Models
for Target-Based Sentiment and Stock
Return

Published in the *International Journal of Data Science and Analytics*

Article Dataset Construction

Dataset Overview

- **Source:** Bloomberg Terminal
- **News Articles:** 1170
- **Target Firm:** Amazon, Netflix, Nvidia, Alphabet

Two-Phase Annotation

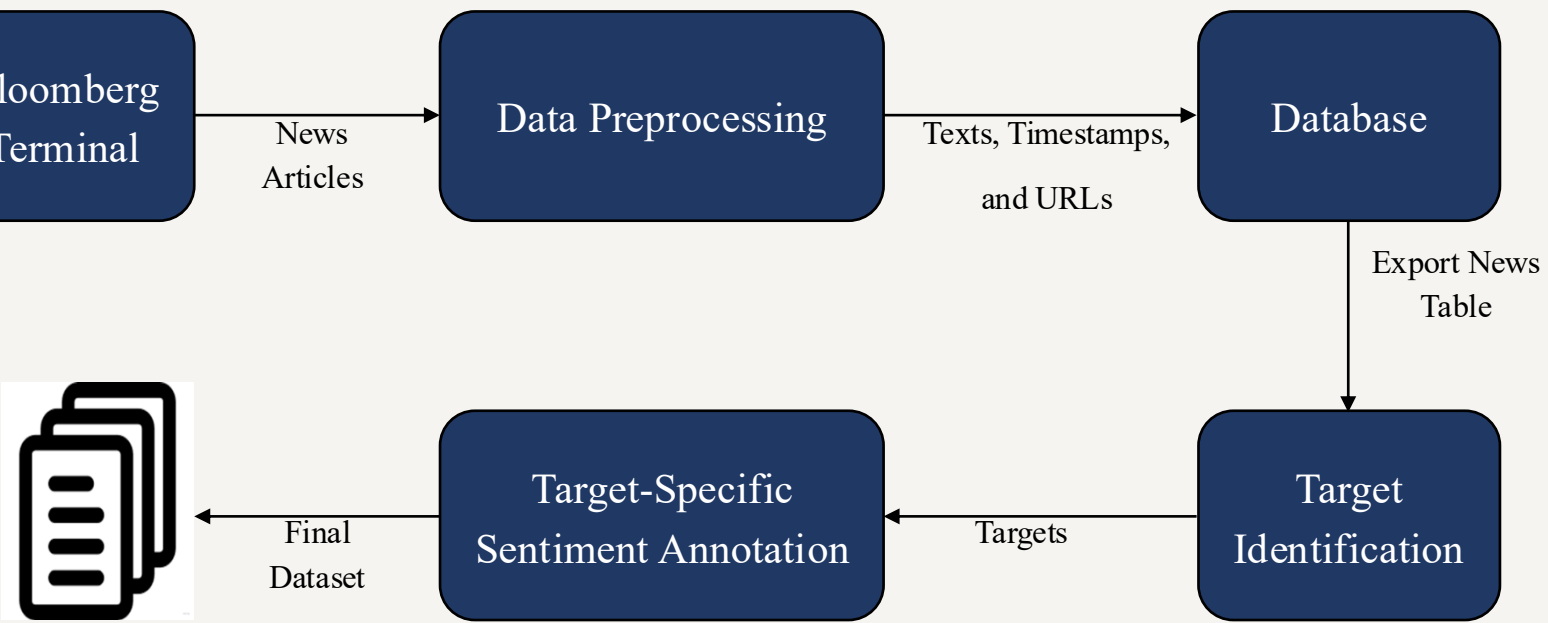
- Target identification: $\alpha = 0.96$
- Sentiment labeling: $\alpha = 0.814$

Final Dataset

- 1334 Target-Specific texts

Data Statistics

Target	No of Texts	Daily Avg.	Text Tokens	Target Mentions
Alphabet	247	2.84	456.85	3.29
Amazon	496	4.82	538.31	5.19
Netflix	233	3.11	245.92	3.18
Nvidia	358	3.81	382.97	3.32
Total	1334	14.58	430.20	3.99



Dataset construction process

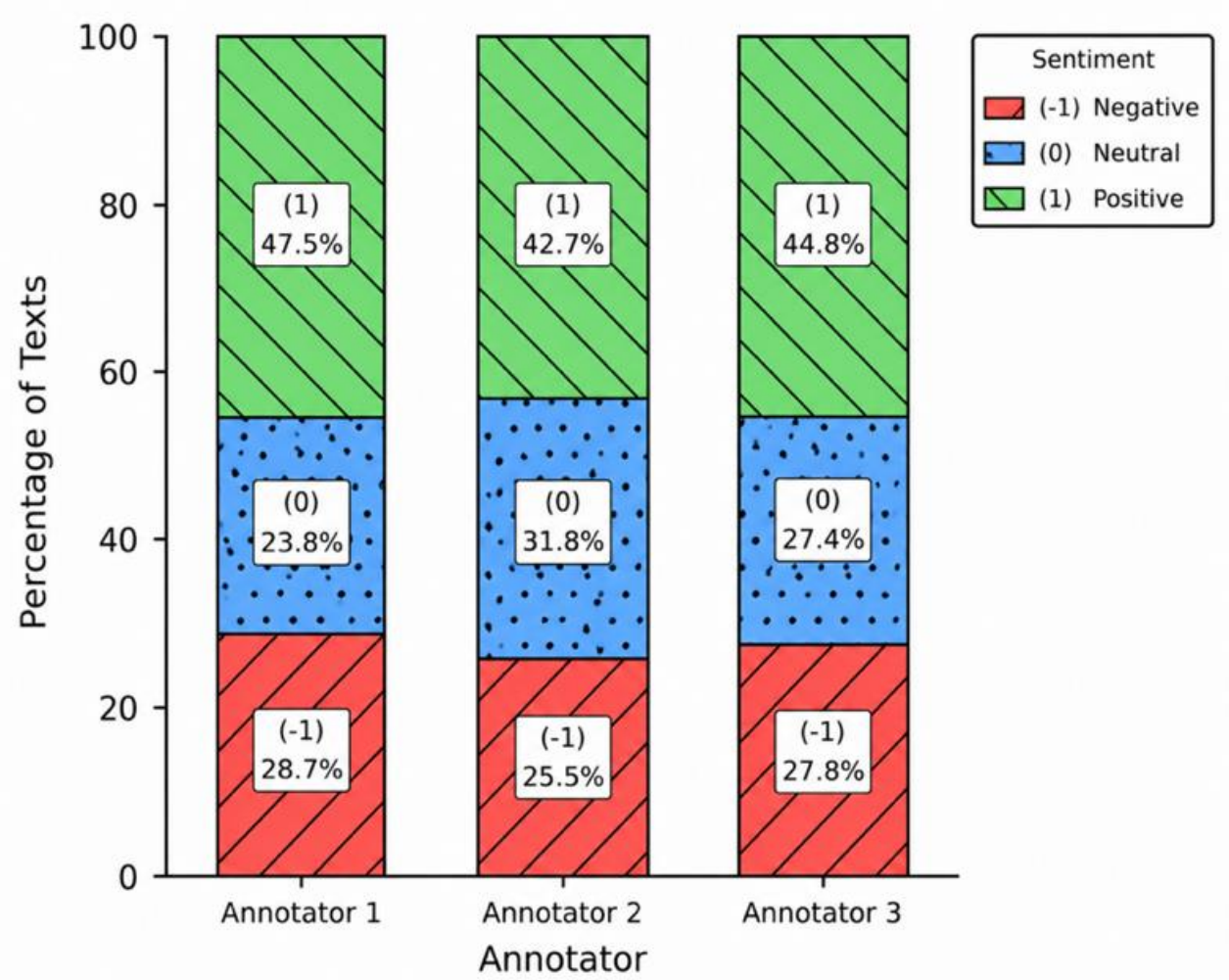
Annotated Examples

Major US technology and internet stocks are lower on Friday, as investors digested a strong jobs report and the implications it could have for Federal Reserve policy. Read more: US Hiring Surges, Bolstering Case for Another Fed Rate Hike The Nasdaq 100 Index fell 0.7% Among megacap stocks: Apple -0.4%, Microsoft -0.2%, Amazon.com -1.1%, Alphabet -0.2%, Nvidia - 0.7% and Meta Platforms -0.7% The Nasdaq 100 is down 7.8% off a July peak, with concerns over Fed policy and rising Treasury yields pressuring valuations However, the weakness in tech has created some opportunities given the sector’s strong earnings prospects, according to UBS “With a rising contribution from AI, better base effects, and pricing power, we expect global tech earnings to turn positive in the third quarter and grow by around 16% for full-year 2024,” writes Solita Marcelli, chief investment officer Americas at UBS Global Wealth Management, who adds that tech “should emerge as the fastest-growing segment within global equities next year” UBS is neutral on the sector, given concerns over valuation, but it writes that “beaten-down parts of the industry provide greater potential upside” The firm sees opportunities in “select semiconductor stocks,” as well as resilient parts of the sector, including the software, internet and services industries.

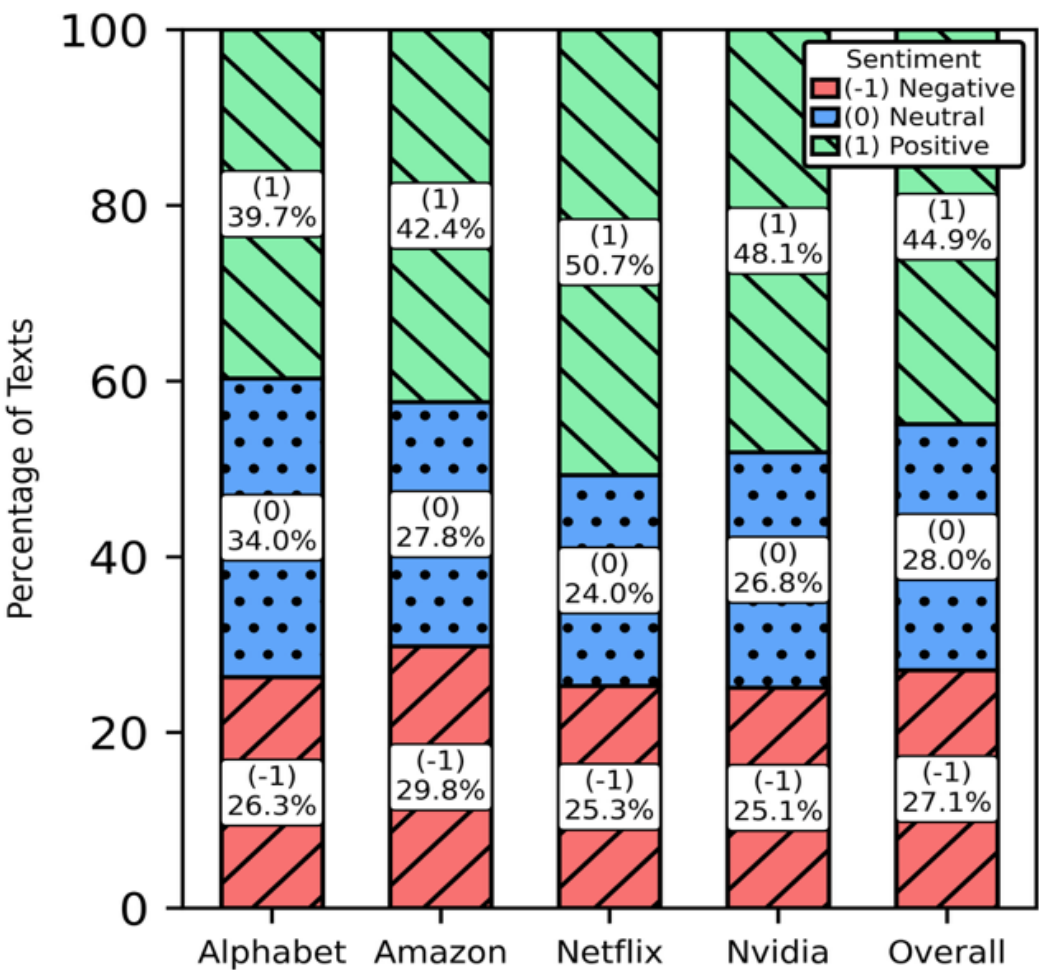
Target	Text	Label
Alphabet	Alphabet Inc. shares tumbled the most in a year on Wednesday after the Google parent reported a smaller than expected profit in cloud computing, raising concerns about its position in a market critical to its future. Ed Ludlow reports.	-1
Amazon	Amazon Japan says it will build its first “sort center” in Japan in Shinagawa, Tokyo, located ~3.5km from Haneda International Airport. Expects to create ~1,000 new jobs. Will handle as many as 750,000 items/day.	1
Netflix	Netflix co-CEO Ted Sarandos says talks with striking actors broke down after the union asked for a “levy” on streaming customers. Sarandos speaks at the first-ever Bloomberg Screentime conference in Los Angeles.	-1
Nvidia	The projected ex-date for Nvidia's dividend moved to Dec. 6 from Nov. 30, according to an updated Bloomberg Dividend Forecast. The new ex-date falls after the Dec. 1 option expiry.	0

(Target, Sentiment):
(Amazon, -1)
(Alphabet, -1)
(Nvidia, -1)

Sentiment Distribution



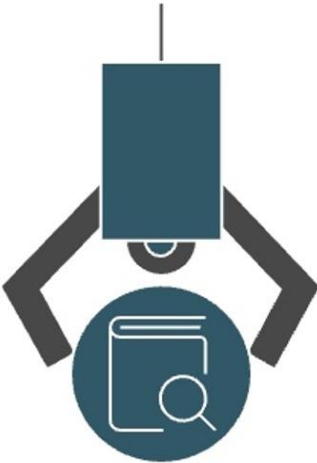
Sentiment labels assigned by annotators



Sentiment distribution across targets

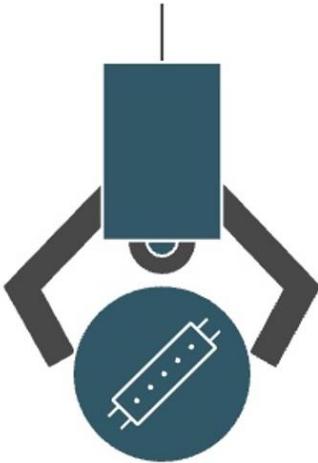


Models Evaluated



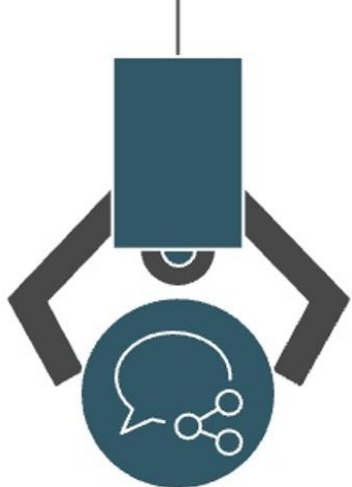
Lexicon-based tools

VADER,
TextBlob



Transformer models

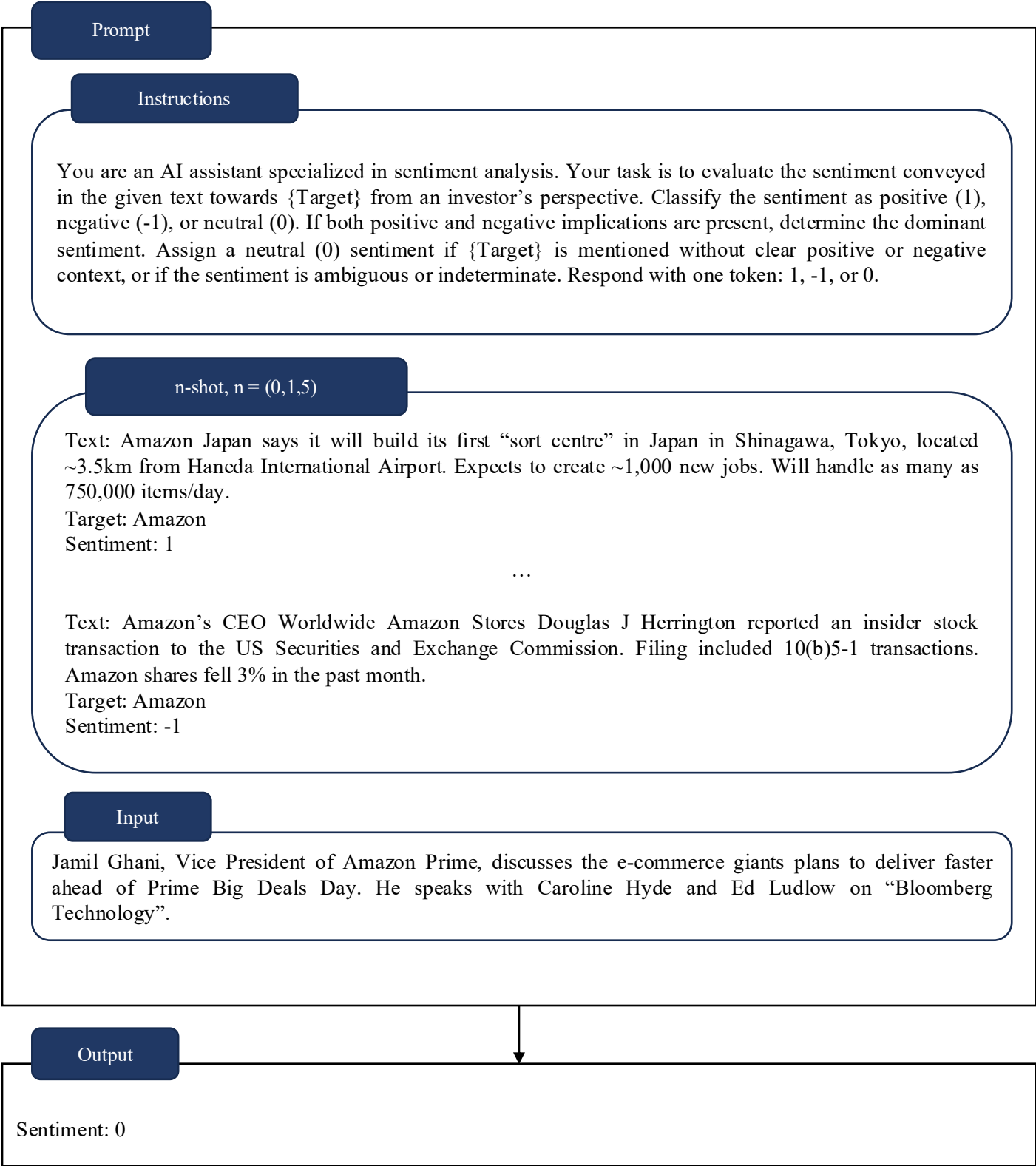
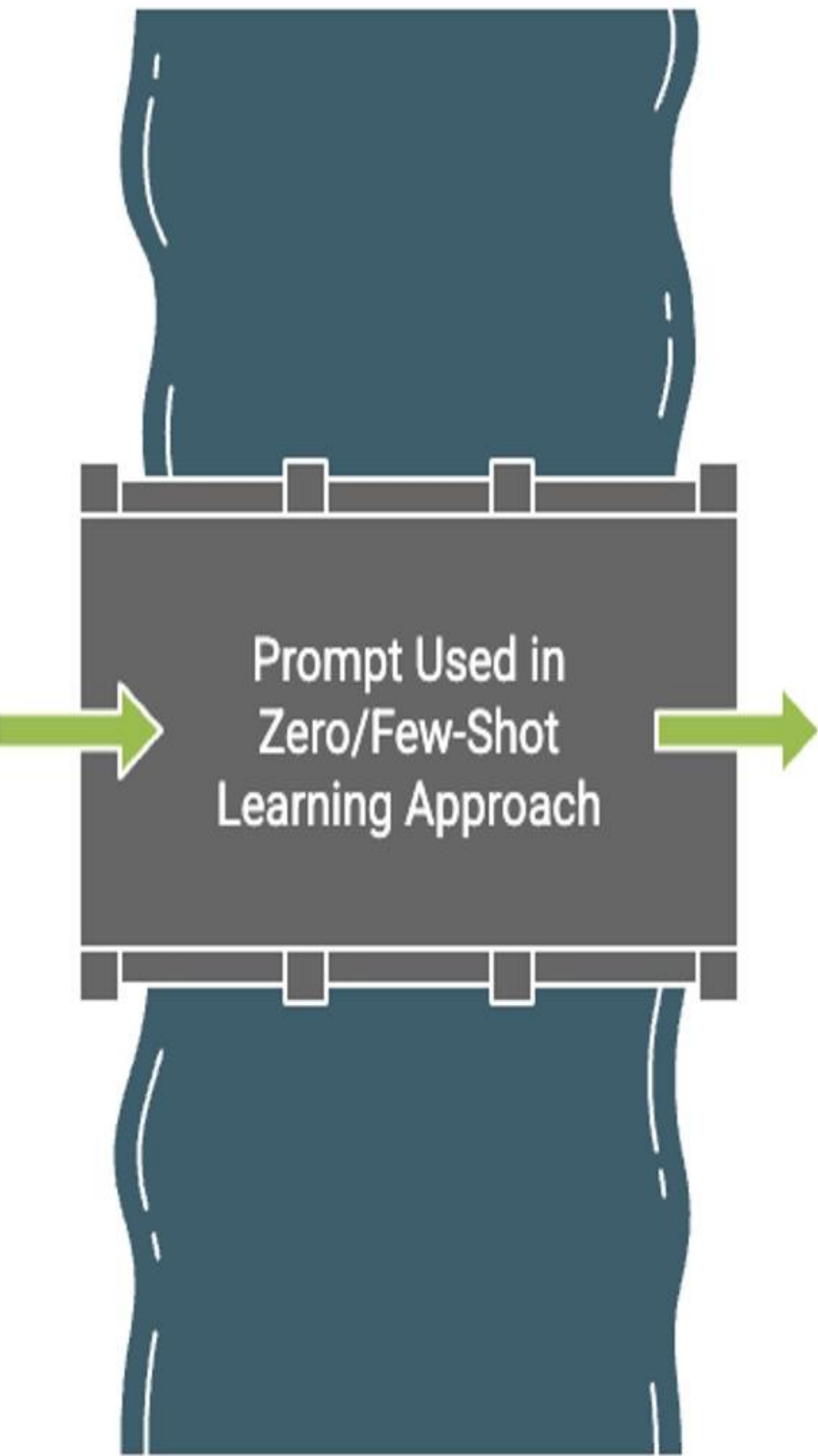
DistilFinRoBERTa,
FinBERT,
FinBERT-Tone,
Deberta-v3-base-absa-v1.1



LLMs

ChatGPT (4o, 4, o1),
DeepSeek (R1),
Llama (3-8B),
Gemma (2-9B, 2-27B)

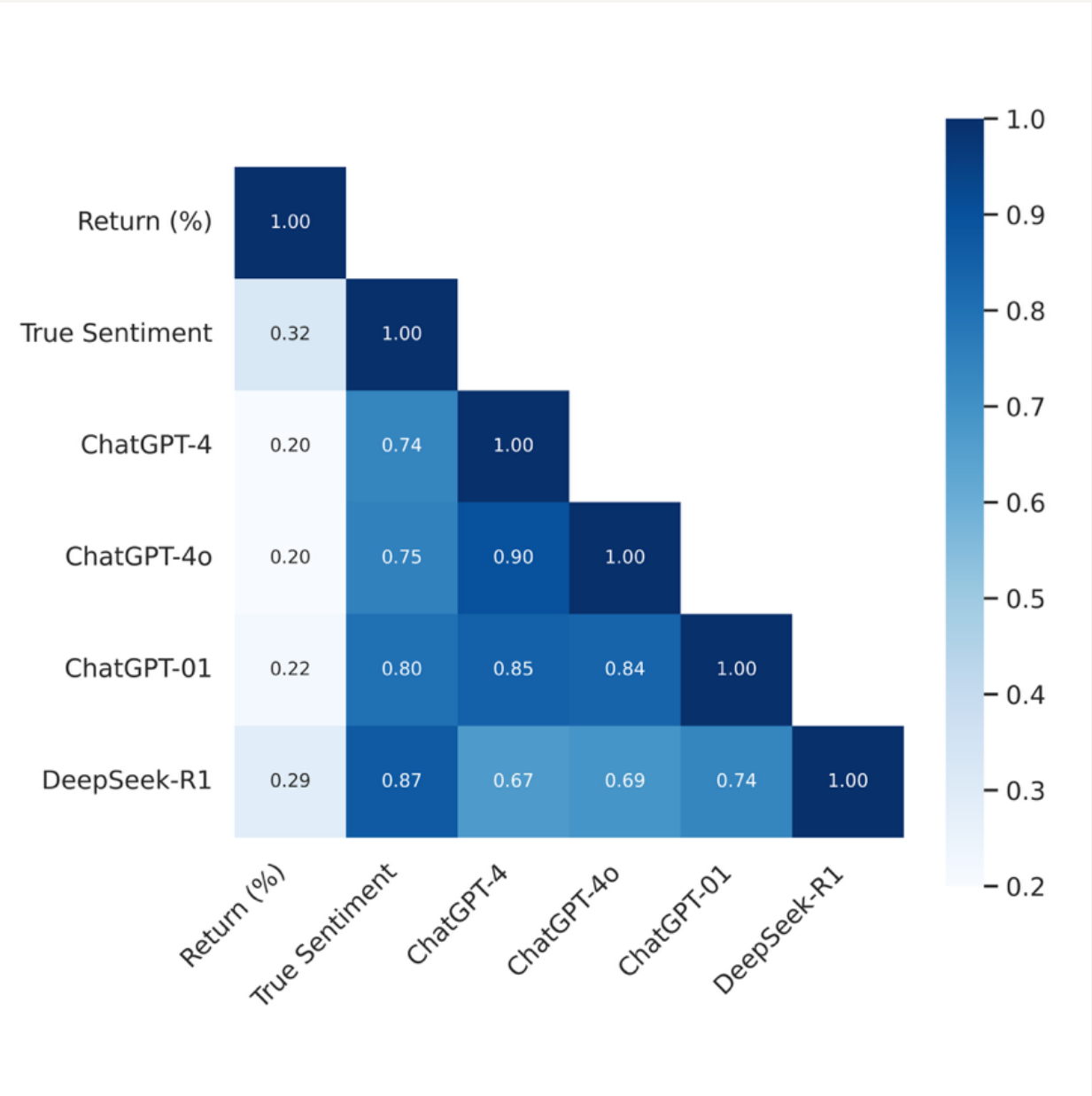
Baseline Models



Performance Outcomes of Target-level Sentiment Classification Across Models

Model	Shot	Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Weighted Precision	Weighted Recall	Weighted F1-Score
<u>TextBlob</u>	-	0.46	0.40	0.39	0.35	0.41	0.46	0.39
VADER	-	0.50	0.48	0.41	0.37	0.48	0.50	0.41
<u>FinBERT</u>	-	0.56	0.53	0.54	0.54	0.58	0.56	0.57
<u>DistilFinRoBERTa</u>	-	0.61	0.56	0.57	0.57	0.61	0.61	0.61
<u>FinBERT-Tone</u>	-	0.63	0.61	0.62	0.62	0.66	0.63	0.63
Deberta-v3-absa-v1.1	-	0.68	0.67	0.67	0.66	0.68	0.68	0.68
Llama 3 8B	0	0.68	0.75	0.62	0.63	0.72	0.68	0.66
Gemma 2 9B	0	0.66	0.69	0.65	0.66	0.71	0.66	0.67
Gemma 2 27B	0	0.69	0.70	0.68	0.69	0.71	0.69	0.70
ChatGPT-4	0	0.79	0.78	0.77	0.77	0.79	0.79	0.79
ChatGPT-4o	0	0.78	0.78	0.77	0.77	0.79	0.78	0.78
ChatGPT-o1	0	0.81	0.81	0.80	0.80	0.82	0.81	0.81
DeepSeek-R1	0	0.82	0.84	0.81	0.81	0.83	0.83	0.82
Llama 3 8B	1	0.52	0.60	0.44	0.44	0.54	0.40	0.43
Gemma 2 9B	1	0.66	0.70	0.66	0.66	0.72	0.66	0.67
Gemma 2 27B	1	0.71	0.72	0.71	0.71	0.73	0.71	0.72
ChatGPT-4	1	0.80	0.79	0.79	0.79	0.81	0.80	0.80
ChatGPT-4o	1	0.81	0.81	0.80	0.80	0.82	0.81	0.81
ChatGPT-o1	1	0.85	0.85	0.84	0.84	0.85	0.85	0.85
DeepSeek-R1	1	0.83	0.84	0.82	0.83	0.83	0.83	0.83
Llama 3 8B	5	0.64	0.69	0.61	0.63	0.64	0.64	0.63
Gemma 2 9B	5	0.65	0.68	0.65	0.65	0.71	0.65	0.66
Gemma 2 27B	5	0.72	0.71	0.71	0.71	0.73	0.72	0.72
ChatGPT-4	5	0.82	0.81	0.80	0.80	0.82	0.82	0.82
ChatGPT-4o	5	0.82	0.82	0.81	0.82	0.83	0.82	0.83
ChatGPT-o1	5	0.86	0.86	0.85	0.86	0.86	0.86	0.86
DeepSeek-R1	5	0.87	0.88	0.86	0.87	0.87	0.87	0.87

Sentiment-Return Correlation

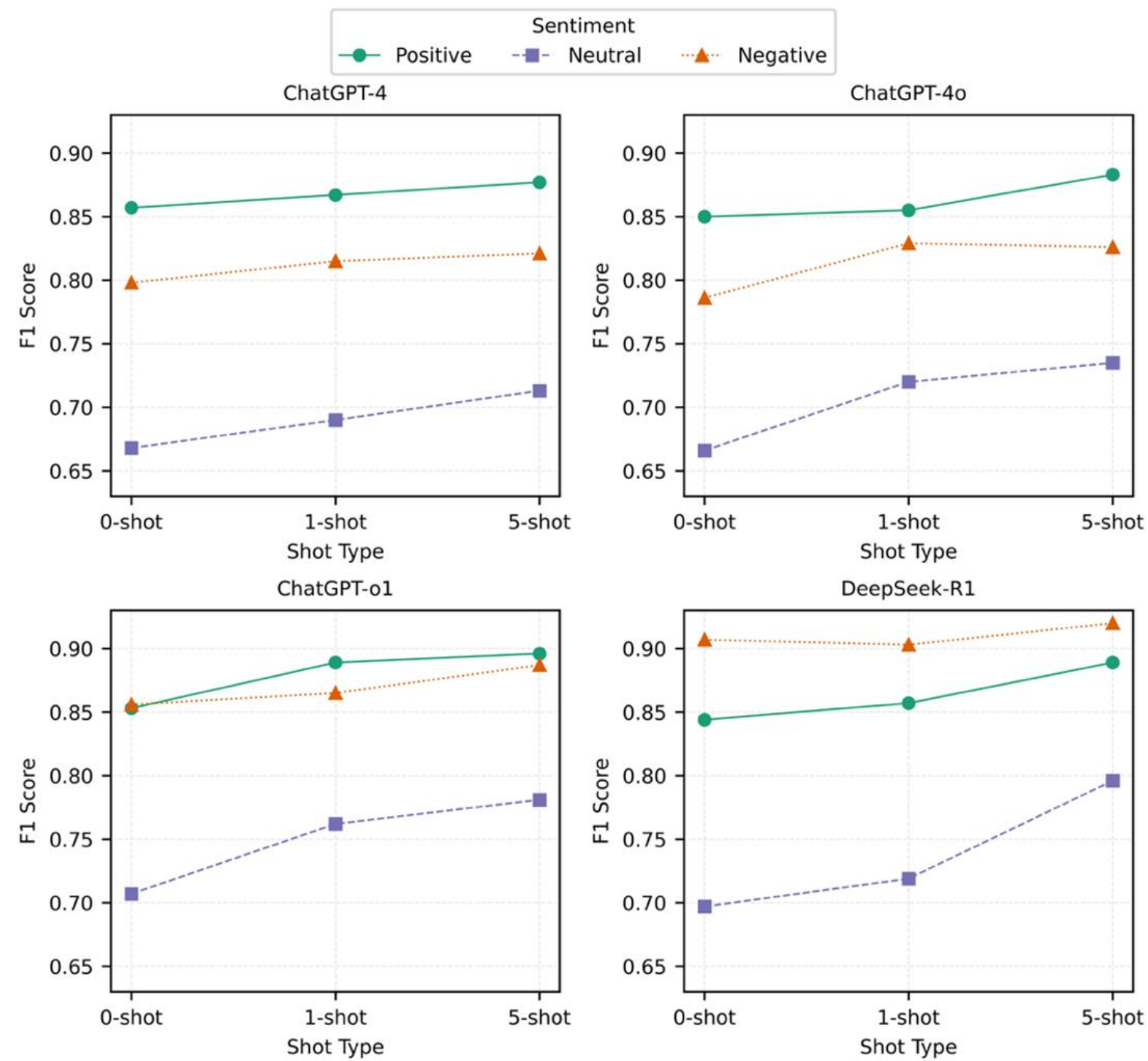


Pearson Correlation: Sentiment vs. Returns

Sentiment-Return Directional Accuracy (%)

Model	Shot	Amazon	Netflix	Nvidia	Alphabet	Overall
ChatGPT-4	0	42.72	42.67	58.51	50.57	48.75
ChatGPT-4o	0	45.63	44.00	55.32	47.13	48.19
ChatGPT-o1	0	49.51	52.00	55.32	42.53	49.86
Deepseek-R1	0	56.31	56.00	64.89	47.13	56.27
ChatGPT-4	1	42.72	46.67	56.38	48.28	48.47
ChatGPT-4o	1	47.57	48.00	50.00	45.98	47.91
ChatGPT-o1	1	44.66	50.67	59.57	47.13	50.42
Deepseek-R1	1	56.31	56.00	69.15	51.72	58.50
ChatGPT-4	5	43.69	46.67	58.51	49.43	49.58
ChatGPT-4o	5	46.60	49.33	57.45	43.68	49.30
ChatGPT-o1	5	51.46	54.67	58.51	43.68	52.09
Deepseek-R1	5	60.19	58.67	71.28	52.87	61.00
True Sentiment	-	66.02	66.67	76.60	66.67	69.08
Naïve Sentiment	-	52.43	48.00	60.64	54.02	54.04

Zero-Shot Versus Few Shot Analysis



Comparison of 0-shot and few-shot performance of generative LLMs across sentiment classes.

Key Findings

- Lexicon-based methods underperform due to static lexicons and document-level analysis.
- Among transformer-based models, DeBERTa-v3-base-ABSA-v1.1 achieves the best performance.
- Generative LLMs (DeepSeek-R1, ChatGPT-o1) consistently outperform lexicon-based methods and discriminative transformers.
- Few-shot prompting benefits larger LLMs but can degrade smaller models (e.g., Llama 3 8B).
- LLM-derived sentiment exhibits a statistically significant positive association with contemporaneous stock returns.

05

Economic Validation

Short-horizon return forecasting, trading performance, and risk-adjusted alpha

Forecasting Risk-Adjusted Stock Returns
with Firm-Specific News Sentiment:
Evidence from Large Language Models

Under Review at the *Journal of Forecasting*

Textual Data

Textual Data Sources

Firm	Headlines	Articles
Amazon	493	441
Netflix	257	202
Nvidia	337	312
Alphabet	189	215
Total	1276	1170

Sentiment Extraction Models

- Discriminative transformer (baseline):
 - **FinBERT**
- Generative LLMs:
 - **ChatGPT-o3**
 - **DeepSeek-R1**
- All models extract {-1, 0, +1} labels at target-level

Prompt

You are an AI assistant specialized in sentiment analysis. Evaluate the sentiment of the given text toward {Firm} from an investor’s perspective. Classify the sentiment as Positive (1), Negative (-1), or Neutral (0). If both positive and negative implications exist, select the dominant sentiment. Respond with one token: 1, -1, or 0.

Text: {}

Output

Sentiment: {}

Prompt used for generative LLMs.

Methodology: Features, Models & Evaluation

FEATURES

Market-Based:

- Lagged returns
- RSI
- SMA

Sentiment:

- Daily avg sentiment
- 3-day momentum & volatility

Controls:

- Firm fixed effects (D_T)

MODELS

Linear:

- Logistic Regression

Nonlinear:

- Random Forest
- XGBoost

EVALUATION

Statistical:

- Accuracy
- F1-score

Trading Metrics:

- Sharpe ratio
- Maximum drawdown
- Win rate

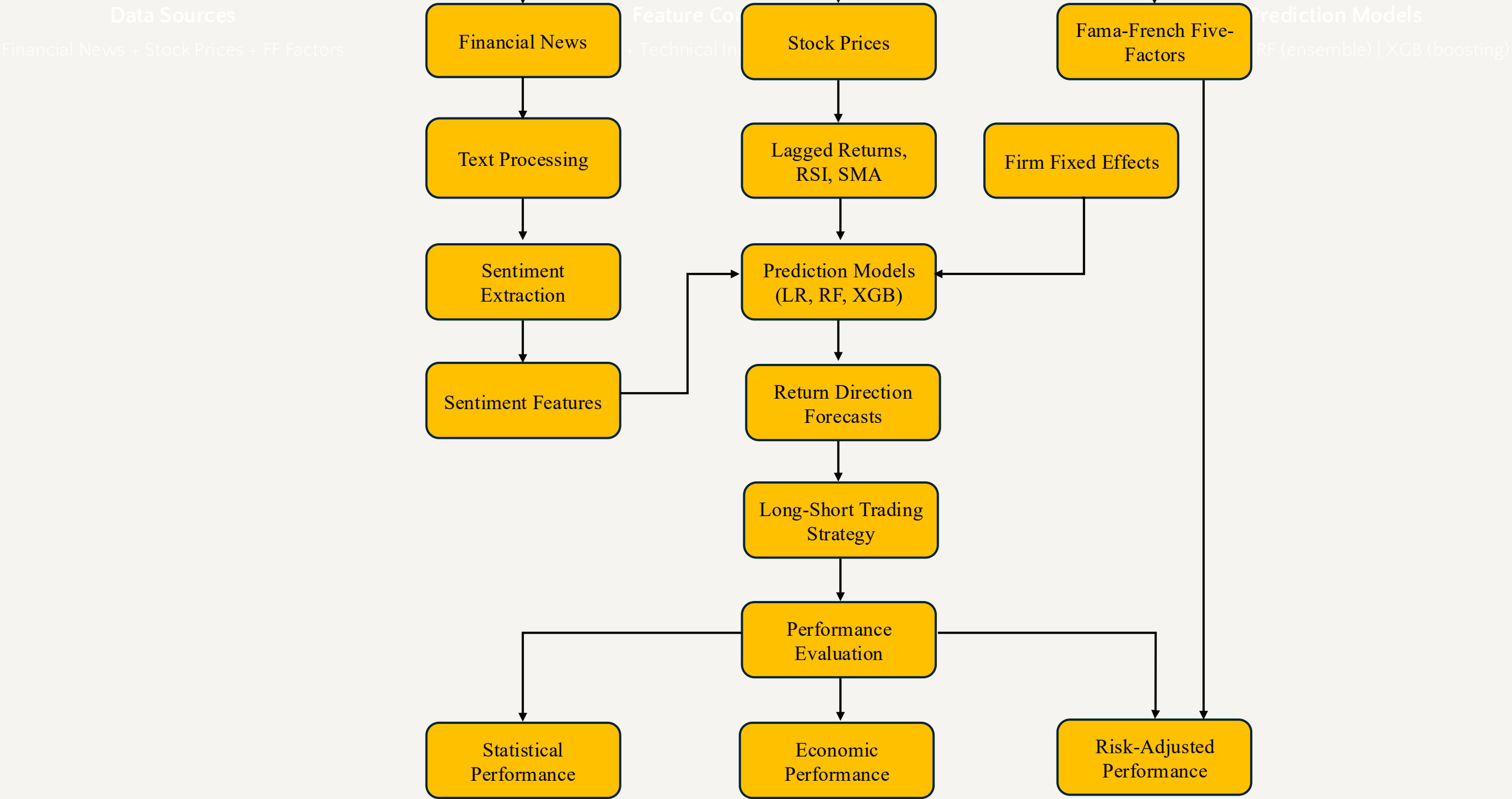
Risk-Adjusted:

- FF5 Alpha

Experimental Setup

- **Forecast Horizons:** 1-, 2-, and 3-day ahead
- **Firms:** Amazon, Netflix, Nvidia, Alphabet
- **Sample:** Sep 2023 - Jan 2024

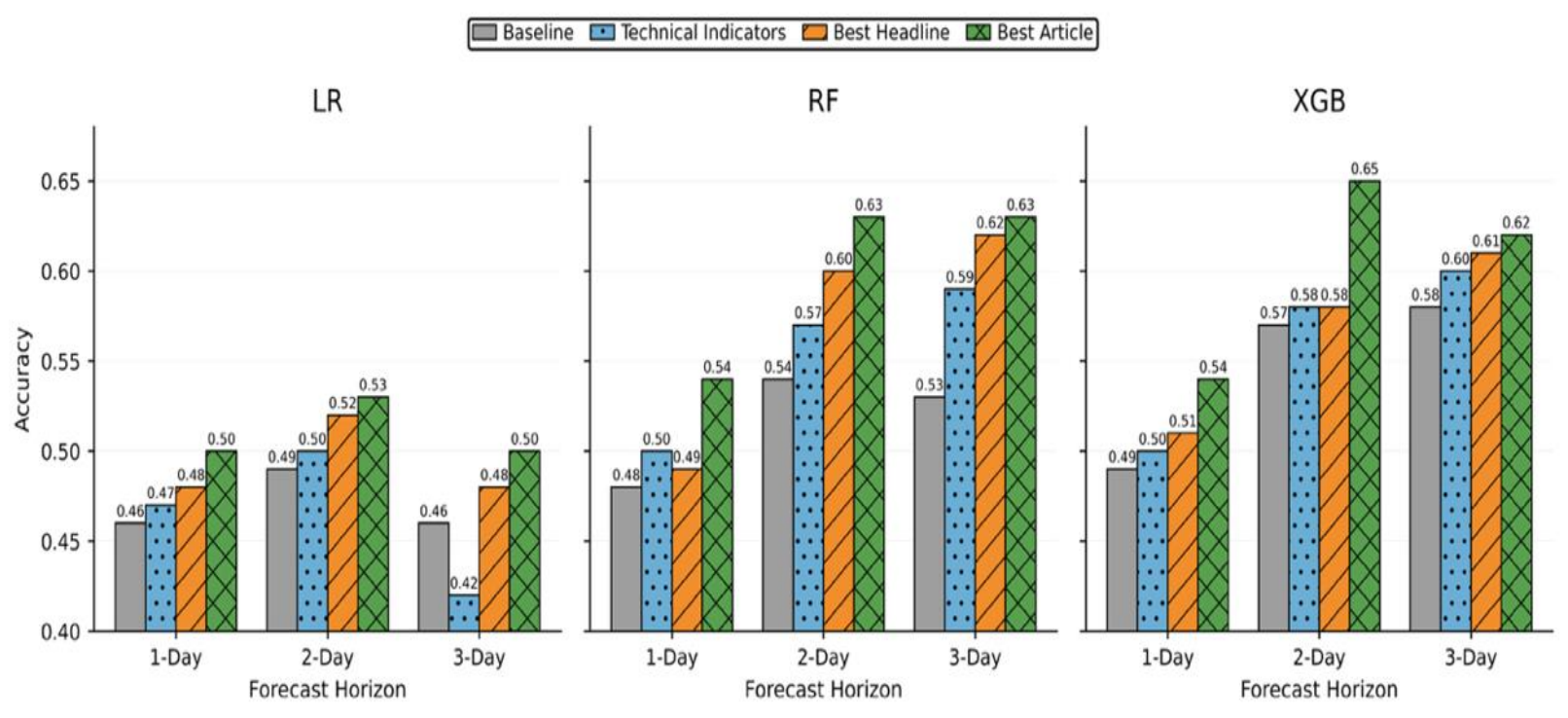
Predictive Framework



Predictive Performance Results

Features	Model	1-day horizon		2-day horizon		3-day horizon	
		Acc.	F1	Acc.	F1	Acc.	F1
Returns + D_T (Baseline)	LR	0.46	0.39	0.49	0.50	0.46	0.41
Returns + D_T (Baseline)	RF	0.48	0.48	0.54	0.63	0.53	0.62
Returns + D_T (Baseline)	XGB	0.49	0.48	0.57	0.67	0.58	0.69
Returns + TIs + D_T	LR	0.47	0.39	0.53	0.53	0.42	0.34
Returns + TIs + D_T	RF	0.50	0.49	0.57	0.67	0.59	0.70
Returns + TIs + D_T	XGB	0.50	0.49	0.58	0.65	0.61	0.71
<u>News headline sentiment</u>							
Returns + TIs + D_T + FinBERT	LR	0.45	0.27	0.50	0.51	0.46	0.43
Returns + TIs + D_T + ChatGPT-o3	LR	0.47	0.29	0.52	0.52	0.47	0.46
Returns + TIs + D_T + DeepSeek-R1	LR	0.48	0.31	0.51	0.52	0.48	0.47
Returns + TIs + D_T + FinBERT	RF	0.47	0.45	0.57	0.68	0.58	0.70
Returns + TIs + D_T + ChatGPT-o3	RF	0.49	0.46	0.60	0.72	0.62	0.72
Returns + TIs + D_T + DeepSeek-R1	RF	0.48	0.50	0.58	0.69	0.60	0.71
Returns + TIs + D_T + FinBERT	XGB	0.47	0.49	0.57	0.64	0.55	0.65
Returns + TIs + D_T + ChatGPT-o3	XGB	0.50	0.50	0.57	0.66	0.56	0.66
Returns + TIs + α_F + DeepSeek-R1	XGB	0.51	0.53	0.58	0.67	0.60	0.68
<u>News article sentiment</u>							
Returns + TIs + D_T + FinBERT	LR	0.46	0.32	0.48	0.51	0.41	0.40
Returns + TIs + D_T + ChatGPT-o3	LR	0.47	0.34	0.50	0.50	0.42	0.41
Returns + TIs + D_T + DeepSeek-R1	LR	0.50	0.40	0.49	0.51	0.50	0.50
Returns + TIs + D_T + FinBERT	RF	0.50	0.52	0.62	0.73	0.60	0.72
Returns + TIs + D_T + ChatGPT-o3	RF	0.54	0.60	0.63	0.74	0.63	0.73
Returns + TIs + D_T + DeepSeek-R1	RF	0.52	0.53	0.62	0.73	0.60	0.73
Returns + TIs + D_T + FinBERT	XGB	0.52	0.53	0.58	0.66	0.56	0.65
Returns + TIs + D_T + ChatGPT-o3	XGB	0.54	0.57	0.58	0.69	0.62	0.73
Returns + TIs + D_T + DeepSeek-R1	XGB	0.53	0.54	0.65	0.74	0.59	0.70

Note: Acc. and F1 denote accuracy and F1-score, respectively.



Incremental predictive performance across feature sets and forecast horizons.

Economic Performance

Features	Model	1-day horizon			2-day horizon			3-day horizon		
		SR	MDD	Win	SR	MDD	Win	SR	MDD	Win
Returns + D_T (Baseline)	LR	-0.08	-19.39	0.48	0.12	-37.35	0.56	-0.08	-52.30	0.46
Returns + D_T (Baseline)	RF	-0.01	-18.34	0.48	0.00	-58.34	0.54	0.02	-69.99	0.53
Returns + D_T (Baseline)	XGB	-0.08	-19.62	0.52	0.00	-63.02	0.57	0.05	-42.62	0.58
Returns + TIs + D_T	LR	-0.12	-19.55	0.45	0.06	-51.47	0.53	-0.07	-65.09	0.42
Returns + TIs + D_T	RF	-0.05	-17.32	0.51	0.16	-29.34	0.56	0.18	-38.07	0.59
Returns + TIs + D_T	XGB	-0.02	-18.17	0.48	0.10	-51.30	0.57	0.22	-35.30	0.61
<u>News headline sentiment</u>										
Returns + TIs + D_T + FinBERT	LR	-0.09	-19.27	0.46	0.11	-51.99	0.53	0.11	-59.68	0.50
Returns + TIs + D_T + ChatGPT-o3	LR	-0.12	-20.05	0.45	0.12	-56.60	0.51	0.04	-56.60	0.47
Returns + TIs + D_T + DeepSeek-R1	LR	-0.10	-18.88	0.46	0.11	-59.43	0.51	0.07	-88.70	0.48
Returns + TIs + D_T + FinBERT	RF	0.00	-18.18	0.50	0.13	-25.63	0.60	0.18	-41.17	0.58
Returns + TIs + D_T + ChatGPT-o3	RF	0.02	-14.44	0.53	0.18	-22.64	0.60	0.21	-31.29	0.61
Returns + TIs + D_T + DeepSeek-R1	RF	-0.07	-18.53	0.48	0.18	-24.62	0.57	0.19	-32.99	0.60
Returns + TIs + D_T + FinBERT	XGB	0.01	-14.16	0.51	0.10	-58.75	0.57	0.13	-31.53	0.61
Returns + TIs + D_T + ChatGPT-o3	XGB	-0.08	-19.90	0.46	0.14	-31.12	0.57	0.13	-36.15	0.56
Returns + TIs + D_T + DeepSeek-R1	XGB	-0.02	-14.70	0.50	0.18	-22.68	0.55	0.20	-37.70	0.60
<u>News article sentiment</u>										
Returns + TIs + D_T + FinBERT	LR	-0.09	-17.08	0.48	0.06	-62.10	0.48	-0.07	-61.20	0.46
Returns + TIs + D_T + ChatGPT-o3	LR	-0.09	-19.09	0.45	0.10	-46.10	0.50	-0.04	-56.83	0.42
Returns + TIs + D_T + DeepSeek-R1	LR	-0.06	-19.27	0.48	0.06	-47.50	0.49	0.06	-58.88	0.50
Returns + TIs + D_T + FinBERT	RF	0.03	-15.48	0.50	0.24	-24.88	0.62	0.24	-32.86	0.60
Returns + TIs + D_T + ChatGPT-o3	RF	0.09	-11.06	0.53	0.21	-23.71	0.63	0.25	-29.82	0.60
Returns + TIs + D_T + DeepSeek-R1	RF	0.09	-11.72	0.51	0.16	-22.89	0.62	0.18	-33.08	0.60
Returns + TIs + D_T + FinBERT	XGB	0.03	-17.08	0.51	0.22	-28.67	0.58	0.19	-36.44	0.56
Returns + TIs + D_T + ChatGPT-o3	XGB	0.02	-15.12	0.53	0.10	-47.57	0.58	0.24	-33.91	0.63
Returns + TIs + D_T + DeepSeek-R1	XGB	0.08	-11.18	0.51	0.24	-18.85	0.65	0.22	-30.48	0.59
Notes: SR, MDD, and W denote the Sharpe ratio, maximum drawdown (%), and win rate (%), respectively.										

Risk-Adjusted Alpha: Fama-French Five-Factor

Features	Model	1-day horizon		2-day horizon		3-day horizon	
		Alpha	t-stat	Alpha	t-stat	Alpha	t-stat
Returns + D_T (Baseline)	LR	-0.002	-2.91	-0.010	0.26	0.006	0.91
Returns + D_T (Baseline)	RF	0.000	0.37	0.002	0.48	0.007	1.20
Returns + D_T (Baseline)	XGB	0.002	2.89	-0.004	-0.79	0.015	2.45
Returns + TIs + D_T	LR	-0.004	5.08	-0.005	-1.01	-0.018	-2.88
Returns + TIs + D_T	RF	-0.002	-2.78	0.017	3.56	0.029	4.74
Returns + TIs + D_T	XGB	-0.001	-1.53	0.007	1.47	0.032	5.14
<u>News headline sentiment</u>							
Returns + TIs + D_T + FinBERT	LR	-0.003	-4.54	0.007	1.36	-0.002	-0.35
Returns + TIs + D_T + ChatGPT-o3	LR	-0.003	-4.37	0.006	1.31	0.009	1.43
Returns + TIs + D_T + DeepSeek-R1	LR	-0.003	-4.54	0.007	1.36	-0.002	-0.35
Returns + TIs + D_T + FinBERT	RF	0.000	0.38	0.013	2.66	0.029	4.74
Returns + TIs + D_T + ChatGPT-o3	RF	-0.003	-3.56	0.019	3.87	0.029	4.76
Returns + TIs + D_T + DeepSeek-R1	RF	0.000	0.38	0.013	2.66	0.029	4.74
Returns + TIs + D_T + FinBERT	XGB	-0.003	-4.17	0.015	3.01	0.027	4.36
Returns + TIs + D_T + ChatGPT-o3	XGB	-0.001	-2.32	0.020	4.14	0.030	4.77
Returns + TIs + D_T + DeepSeek-R1	XGB	-0.003	-4.17	0.015	3.01	0.027	4.36
<u>News article sentiment</u>							
Returns + TIs + D_T + FinBERT	LR	-0.003	-4.30	0.004	0.83	-0.015	-2.40
Returns + TIs + D_T + ChatGPT-o3	LR	-0.003	-3.58	0.005	-1.07	0.002	0.33
Returns + TIs + D_T + DeepSeek-R1	LR	-0.003	-4.54	0.007	1.36	-0.002	-0.35
Returns + TIs + D_T + FinBERT	RF	0.001	-1.54	0.020	4.10	-0.033	5.27
Returns + TIs + D_T + ChatGPT-o3	RF	0.003	3.17	0.021	4.28	0.030	4.76
Returns + TIs + D_T + DeepSeek-R1	RF	0.000	0.38	0.013	2.66	0.029	4.74
Returns + TIs + D_T + FinBERT	XGB	0.000	-0.65	0.017	3.43	0.028	4.49
Returns + TIs + D_T + ChatGPT-o3	XGB	0.003	3.57	0.026	5.52	0.031	4.98
Returns + TIs + D_T + DeepSeek-R1	XGB	-0.003	-4.17	0.015	3.01	0.027	4.36
Note: Alpha values with t-stat ≥ 1.96, indicating statistical significance at the 5% level, are shown in bold.							

Key Findings

- Return-only models provide limited predictive value.
- Technical indicators substantially improve nonlinear forecasting models.
- Article-level sentiment combined with nonlinear models delivers the strongest predictive and economic performance.
- Generative LLMs outperform FinBERT across predictive and economic evaluations.
- Predictive and economic gains strengthen over multi-day horizons, consistent with gradual information diffusion.

06

Conclusion & Future Research Directions

Summary of findings, limitations, and future directions

Answers to Research Questions

RQ	Question	Answer
1	Can LLMs reliably extract firm-level sentiment?	Yes. LLMs reliably extract firm-level sentiment and achieve competitive performance without task-specific fine-tuning.
2	Does few-shot prompting improve accuracy?	Conditionally yes. Few-shot prompting benefits larger LLMs but can degrade smaller models.
3	Are LLM sentiment signals associated with returns?	Yes. LLM-derived sentiment signals are positively associated with short-horizon stock returns.
4	Effect of textual depth and model choice?	Yes. Article-level sentiment and nonlinear models provide the strongest predictive performance.
5	Economic significance in trading settings?	Yes. Article-level LLM sentiment in nonlinear frameworks generates statistically and economically significant risk-adjusted alpha.

Limitations & Future Research

LIMITATIONS

- Limited sample: Four large-cap U.S. technology firms
- Short sample period: Sep 2023 - Jan 2024
- No transaction costs or market impact
- Proprietary LLM reproducibility concerns
- No causal identification
- English-language news only
- Short-horizon forecasting

FUTURE DIRECTIONS

- Broader equity universes and international markets
- Multilingual financial sentiment analysis
- Regime-dependent asset pricing (crisis vs. normal periods)
- Transaction costs and liquidity constraints
- Multi-source narratives (earnings calls, filings, social media)
- Intraday and longer-horizon forecasting

Conclusions

- Firm-level narrative sentiment extracted by modern LLMs provides economically meaningful information that improves short-horizon return prediction and generates statistically significant risk-adjusted abnormal returns.
- Key Enablers of Predictive and Economic Performance:

1

Target-Level Measurement

Firm-specific signals

2

Nonlinear Models

RF & XGBoost

3

Article-Level Information

Articles > Headlines

4

Economic Validation

Fama-French five-factor alpha

Core message: Firm-level narrative sentiment becomes economically valuable only when combined with accurate measurement, appropriate modelling, and rigorous economic validation.

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Thank You

Questions & Discussion

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3.1. Headline Dataset

Inter-Annotator Agreement

Krippendorff's Alpha = **0.8188**

Pairwise Cohen's Kappa:

- Annotator 1 vs. 2: 0.8234
- Annotator 1 vs. 3: 0.8012
- Annotator 2 vs. 3: 0.8319

All annotators held CEFR C1 proficiency. Annotation guidelines included 5+ example cases.

4.2. Examples

Pioneer Disciplined Growth Fund/New boosted its holdings in Nike Inc. Class B and reduced its holdings in Amazon.com Inc. in July, according to data analyzed by Bloomberg. The fund's disclosed holdings rose 1.5% in value to \$1.87 billion. Ashesh Savla and Craig Sterling manage the fund. Highlights: Amazon.com Inc. was the biggest investment, representing 4.2% of disclosed assets. Nine increased holdings, including: Nike Class B: 260,318 shares, up 142% to 443,370 valued at \$48.9 million Danaher Corp.: 64,609, up 42% to 218,835 valued at \$55.8 million Johnson Controls International Plc: 188,544, up 38% to 684,368 valued at \$47.6 million Aptiv Plc: 61,569, up 22% to 343,749 valued at \$37.6 million Keysight Technologies Inc.: 49,291, up 21% to 282,744 valued at \$45.5 million. Eight decreased holdings, including: Amazon: 315,442 shares, down 35% to 584,421 valued at \$78.1 million Alphabet Inc. Class A: 285,536, down 38% to 469,311 valued at \$62.3 million Charles Schwab Corp.: 335,941, down 54% to 282,424 valued at \$18.7 million Adobe Inc.: 38,771, down 34% to 75,220 valued at \$41.1 million Digital Realty Trust Inc.: 109,146, down 29% to 273,469 valued at \$34.1 million. Top holdings: Amazon: down 315,442, to 584,421 shares valued at \$78.1 million AbbVie Inc.: unchanged at 516,602 valued at \$77.3 million Home Depot Inc.: unchanged at 212,506 valued at \$70.9 million Alphabet Inc. Class A: down 285,536, to 469,311 valued at \$62.3 million Microchip Technology Inc.: unchanged at 610,586 valued at \$57.4 million. Pioneer Disciplined Growth returned 19% in the past year, underperforming the 22% return of its benchmark, Russell 1000 Growth Index. The fund returned 14% on an annualized basis over the past five years.

(Target, Sentiment):
(Amazon, -1)
(Alphabet, -1)

Major US technology and internet stocks are lower on Friday, as investors digested a strong jobs report and the implications it could have for Federal Reserve policy. Read more: US Hiring Surges, Bolstering Case for Another Fed Rate Hike The Nasdaq 100 Index fell 0.7% Among megacap stocks: Apple -0.4%, Microsoft -0.2%, Amazon.com -1.1%, Alphabet -0.2%, Nvidia - 0.7% and Meta Platforms - 0.7% The Nasdaq 100 is down 7.8% off a July peak, with concerns over Fed policy and rising Treasury yields pressuring valuations However, the weakness in tech has created some opportunities given the sector’s strong earnings prospects, according to UBS “With a rising contribution from AI, better base effects, and pricing power, we expect global tech earnings to turn positive in the third quarter and grow by around 16% for full-year 2024,” writes Solita Marcelli, chief investment officer Americas at UBS Global Wealth Management, who adds that tech “should emerge as the fastest-growing segment within global equities next year” UBS is neutral on the sector, given concerns over valuation, but it writes that “beaten-down parts of the industry provide greater potential upside” The firm sees opportunities in “select semiconductor stocks,” as well as resilient parts of the sector, including the software, internet and services industries.

(Target, Sentiment):
(Amazon, -1)
(Nvidia, -1)
(Alphabet, -1)

4.1. Article Dataset

Correlation between predicted sentiments and stock returns for best-performing generative LLMs.

Model	Sho t	Amazon	Netflix	Nvidia	Alphabet	Overall
ChatGPT-4	0	0.05	0.30	0.26	0.22	0.21
ChatGPT-4o	0	0.03	0.31	0.18	0.18	0.18
ChatGPT-o1	0	0.03	0.32	0.24	0.18	0.21
DeepSeek-R1	0	0.11	0.29	0.35	0.21	0.25
ChatGPT-4	1	0.03	0.28	0.23	0.21	0.19
ChatGPT-4o	1	-0.01	0.33	0.18	0.19	0.19
ChatGPT-o1	1	-0.04	0.31	0.27	0.32	0.21
DeepSeek-R1	1	0.11	0.31	0.36	0.26	0.27
ChatGPT-4	5	0.03	0.29	0.24	0.20	0.20
ChatGPT-4o	5	0.05	0.32	0.21	0.19	0.20
ChatGPT-o1	5	0.05	0.32	0.24	0.20	0.22
DeepSeek-R1	5	0.13	0.33	0.39	0.26	0.29
True Sentiment	-	0.09	0.35	0.46	0.36	0.32

APPENDIX D

5.1. Descriptive Statistics

Variable	Mean	Std Dev	Min	Max	N
MKT (%)	0.12	0.78	-1.57	2.18	302
SMB (%)	-0.03	0.90	-1.93	3.23	302
HML (%)	0.01	0.61	-1.27	2.03	302
RMW (%)	-0.04	0.54	-1.27	1.49	302
CMA (%)	-0.04	0.32	-0.97	0.77	302
RF (%)	0.02	0.00	0.02	0.02	302
1-Day-Ahead Return (%)	0.41	2.16	-9.51	16.05	302
2-Day-Ahead Return (%)	1.02	13.33	-78.76	136.77	300
3-Day-Ahead Return (%)	1.55	16.37	-78.80	138.83	299
1-Day Lagged Return (%)	0.00	0.02	-0.10	0.16	302
SMA	288.27	163.36	126.72	588.71	302
RSI	57.56	17.86	6.51	96.90	302
FinBERT Headline Sentiment	0.04	0.55	-1.00	1.00	302
FinBERT Headline Sentiment Momentum	0.04	0.32	-0.85	0.77	302
FinBERT Headline Sentiment Volatility	0.47	0.29	0.00	1.15	302
DeepSeek-R1 Headline Sentiment	0.21	0.65	-1.00	1.00	302
DeepSeek-R1 Headline Sentiment Momentum	0.21	0.40	-1.00	1.00	302
DeepSeek-R1 Headline Sentiment Volatility	0.54	0.30	0.00	1.15	302
ChatGPT-o3 Headline Sentiment	0.18	0.59	-1.00	1.00	302
ChatGPT-o3 Headline Sentiment Momentum	0.18	0.37	-0.79	1.00	302
ChatGPT-o3 Headline Sentiment Volatility	0.49	0.28	0.00	1.15	302
FinBERT Article Sentiment	-0.09	0.92	-7.00	5.00	302
FinBERT Article Sentiment Momentum	-0.10	0.52	-2.67	1.67	302
FinBERT Article Sentiment Volatility	0.66	0.64	0.00	6.00	302
DeepSeek-R1 Article Sentiment	0.22	0.65	-2.00	2.00	302
DeepSeek-R1 Article Sentiment Momentum	0.22	0.41	-0.90	1.33	302
DeepSeek-R1 Article Sentiment Volatility	0.50	0.35	0.00	2.08	302
ChatGPT-o3 Article Sentiment	0.25	0.72	-2.00	3.00	302
ChatGPT-o3 Article Sentiment Momentum	0.24	0.44	-1.04	1.67	302
ChatGPT-o3 Article Sentiment Volatility	0.56	0.40	0.00	2.08	302

Notes: MKT is the excess return on the market portfolio. SMB (Small Minus Big), HML (High Minus Low), RMW (Robust Minus Weak), and CMA (Conservative Minus Aggressive) are the size, value, profitability, and investment factors, respectively, from the Fama-French five-factor model. RF denotes the daily risk-free rate. RSI refers to the 14-day Relative Strength Index, and SMA to the 10-day Simple Moving Average.

APPENDIX E

5.2. Key Equations

- Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$, $F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$,

where Precision = $\frac{TP}{TP+FP}$ and Recall = $\frac{TP}{TP+FN}$.

- Strategy returns are:

$$r_{d,h,T}^{strat} = \begin{cases} R_{d+h,T}, & \text{if } \hat{y}_{d,h,T} = 1, \\ -R_{d+h,T}, & \text{if } \hat{y}_{d,h,T} = 0. \end{cases}$$

- Sharpe Ratio = $\frac{E[r_{d,h,T}^{strat}]}{\sigma(r_{d,h,T}^{strat})} \sqrt{252}$,

- MDD (%) = $\max_t \left(\frac{\text{Peak}_t - \text{Through}_t}{\text{Peak}_t} \right) \times 100$,

- Win Rate (%) = $\frac{\sum 1\{r_{d,h,T}^{strat} > 0\}}{N} \times 100$.

- Excess returns are defined as:

$$R_{d,h,T}^{\text{excess}} = R_{d,h,T}^{\text{strat}} - R_{f,d}$$

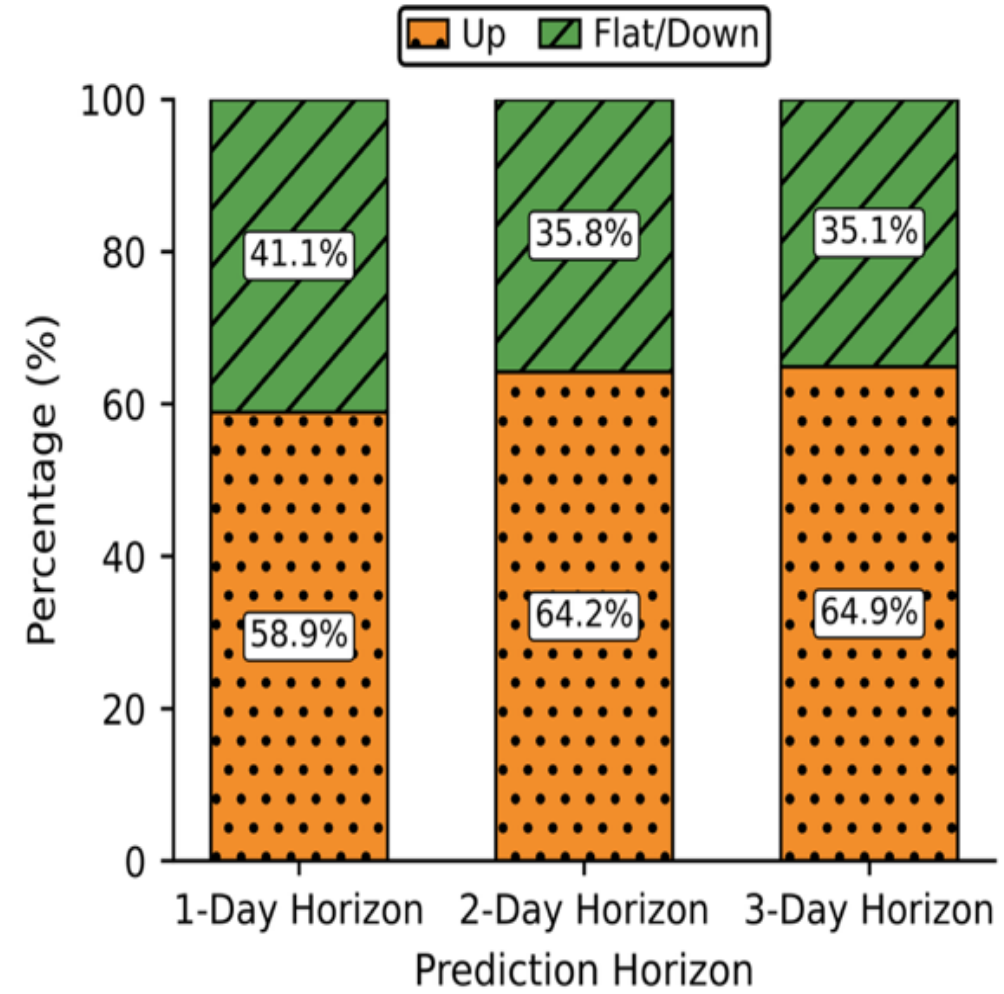
- Fama-French five-factor model is estimated as:

$$R_{d,h,T}^{\text{excess}} = \alpha + \beta_{\text{MKT}}(R_{\text{Mkt},d} - R_{f,d}) + \beta_{\text{SMB}}\text{SMB}_d + \beta_{\text{HML}}\text{HML}_d + \beta_{\text{RMW}}\text{RMW}_d + \beta_{\text{CMA}}\text{CMA}_d + \varepsilon_{d,h,T} \quad (5.19)$$

- Daily sentiment: $\bar{S}_{d,T} = \frac{1}{N_{d,T}} \sum_{j=1}^{N_{d,T}} S_{j,T}$
- The h -day-ahead return: $R_{d+h,T} = \frac{P_{d+h,T} - P_{d,T}}{P_{d,T}} \times 100$, $h \in \{1, 2, 3\}$
- $M_{d,T} = \frac{1}{3} \sum_{k=0}^2 \bar{S}_{d-k,T}$
- $Vol_{d,T} = \sqrt{\frac{1}{3} \sum_{k=0}^2 (\bar{S}_{d-k,T} - M_{d,T})^2}$
- Binary prediction target is:

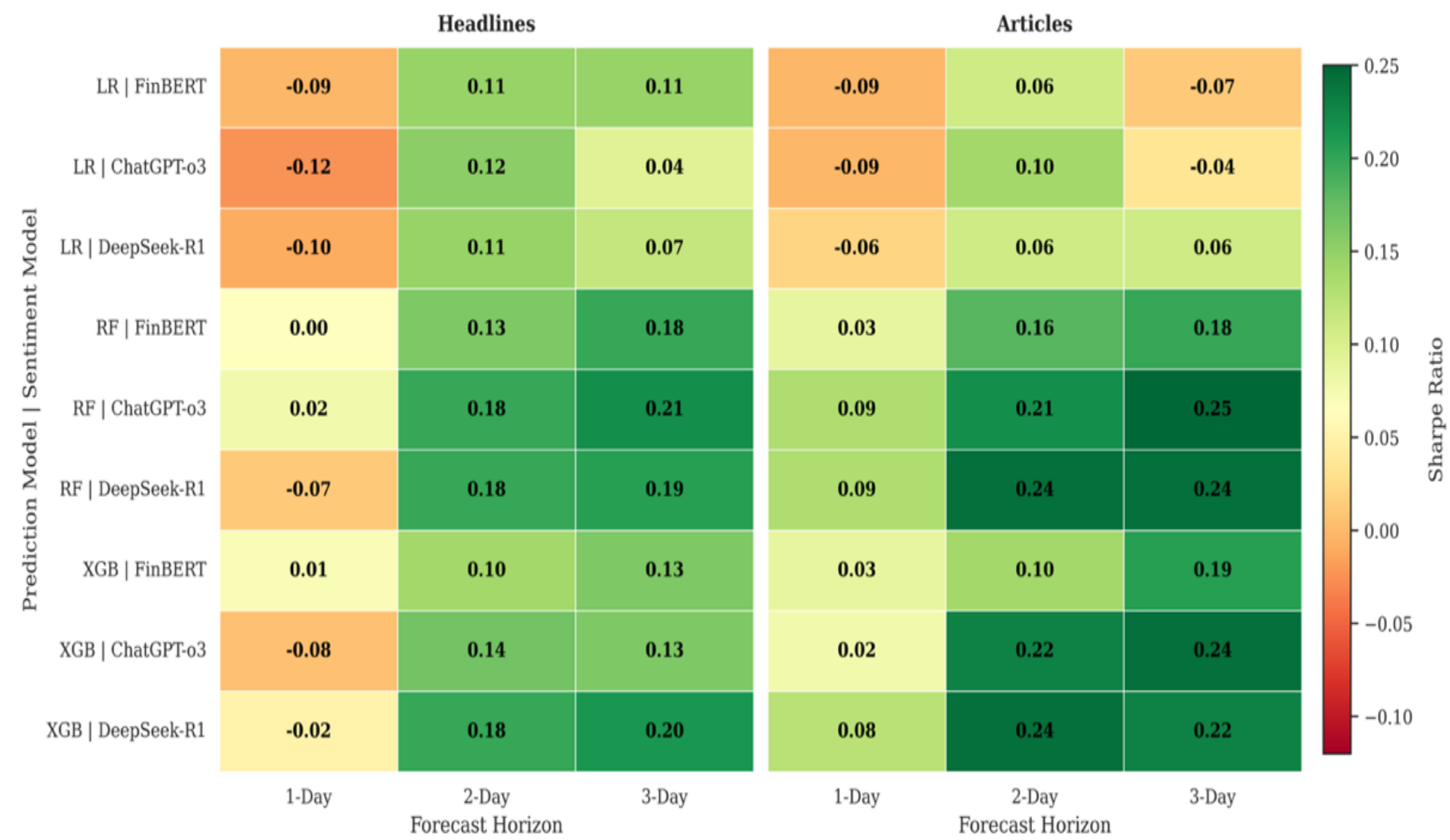
$$y_{d,h,T} = \begin{cases} 1, & \text{if } R_{d+h,T} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

5.3. Distribution of Return Directions



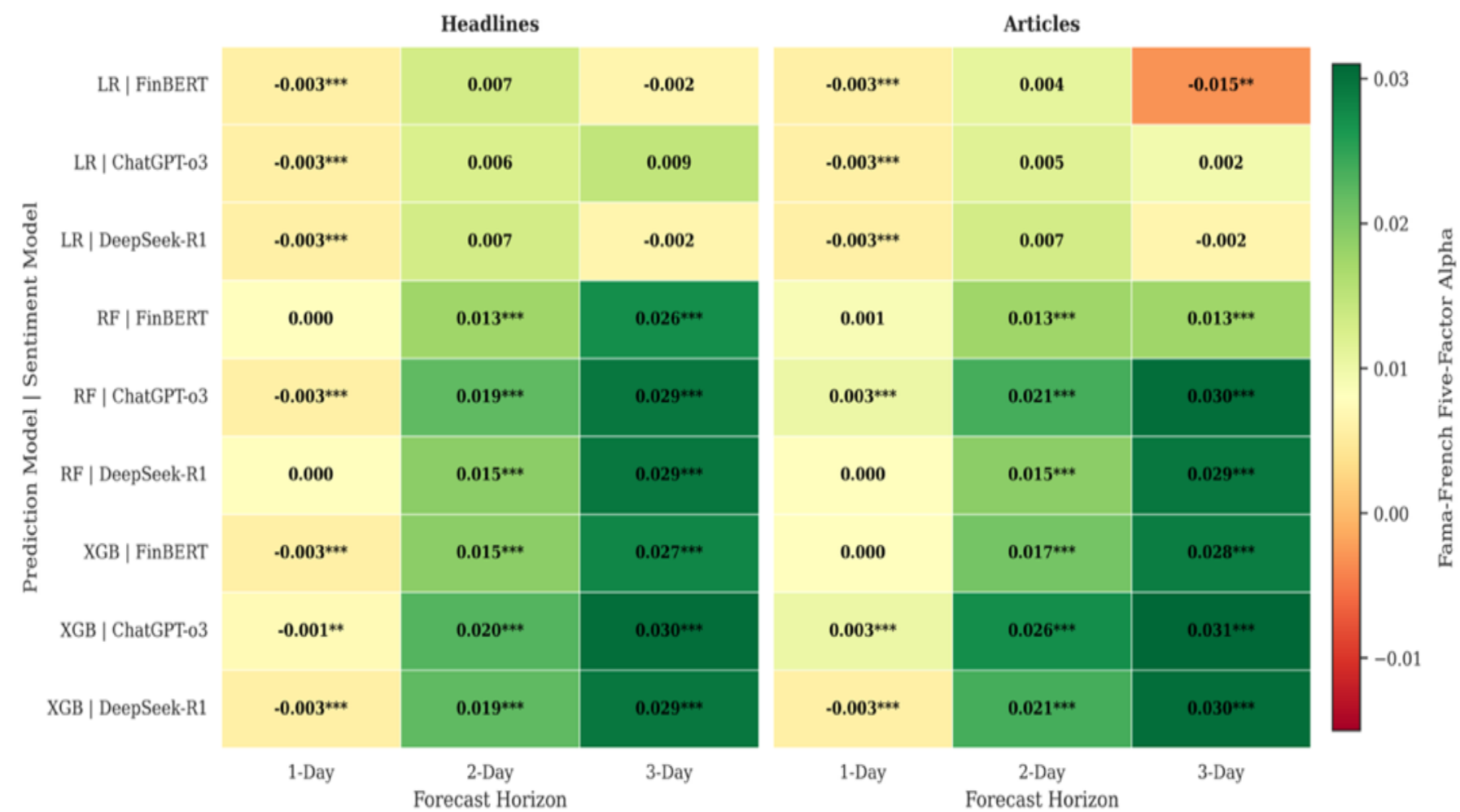
Distribution of return directions across forecast horizons.

5.4. Sharpe Ratios



Sharpe ratios across models and forecast horizons: headline- versus article-level sentiment.

5.5. Fama-French Five-Factor Alphas



Fama-French alphas across models and forecast horizons: headline- versus article-level sentiment.

5.6. Data Availability Statement

- **Dataset 1: Target-Level Financial Sentiment in News Headlines**

The manually annotated dataset of firm-specific financial news headlines developed in this dissertation is publicly available at:

https://github.com/iftikharm895/Target-Level_Financial_Sentiment_Analysis

The repository provides the URLs of the news headlines along with the corresponding manual sentiment annotations.

- **Dataset 2: Target-Based Financial Sentiment in Full-Length News Articles**

The manually annotated dataset of firm-specific full-length financial news articles developed in this dissertation is publicly available at:

https://github.com/iftikharm895/Target-Based_Sentiment_Analysis_in_Financial_News

The repository includes the URLs of the news articles together with the associated manual sentiment annotations.

Due to Bloomberg copyright restrictions, the full text of news headlines and articles cannot be publicly distributed. Researchers with access to a Bloomberg Terminal can reconstruct the datasets using the provided URLs.

